

Tian LIANG

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OBJECTIVE

Aspiring to contribute as a *Research Assistant* in remote sensing and geospatial analysis, with a strong foundation in satellite image processing, UAV-based data acquisition, and 3D modeling. Capable of deep integration of CV large models for the entire process of remote sensing image processing, covering the entire chain of technologies including image enhancement, object detection, and semantic segmentation. Also mastered the innovative integration of SAR, LiDAR, and multispectral image multimodal data, a cross-sensor feature joint modeling framework is constructed, significantly enhancing the intelligent interpretation accuracy of complex ground objects and the generalization ability of scenes.

EDUCATION

Southwest Jiaotong University

Bachelor of Science in Remote Sensing, Sep 2023 – Jun. 2027 (Expected)

- Core Courses: *Surveying Programming Techniques, Fundamentals of Geodesy, Introduction to Cartography, Digital Terrain Surveying, Digital Image Processing, Digital Photogrammetry*
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RESEARCH EXPERIENCE

□ Monitoring the Dynamic Evolution of Glaciers and Glacial Lakes in the Southeastern Tibetan Plateau Using Time-Series Optical and SAR Imagery

Research Assistant | Jan 2024 – Dec 2025

Supervisor: Dr. Liu Guoxiang

Affiliation: National and Local Joint Engineering Laboratory for Spatial Information Technology for Safe Operation of High-Speed Railways

- Focused on remote sensing-based monitoring of glacier and glacial lake evolution under extreme conditions (high elevation, frequent cloud cover).
- Developed theoretical frameworks and algorithms to improve automation and reduce subjective interference in glacier dynamics monitoring.
- Conducted large-scale spatiotemporal analysis using high-resolution optical (e.g., GF-2, GF-7) and multiband SAR images collected over the past decade.
- Designed and implemented deep learning models to enhance the precision of glacial lake boundary identification.
- Investigated key topics including glacier boundary extraction, glacial lake change detection, SAR-based glacier flow velocity monitoring, and mass balance estimation.

□ Extraction and Intelligent Interpretation of Geological Features in Complex Mountainous Terrain Using Multi-Source Remote Sensing

Research Assistant | Dec 2023 – Jan 2024

Supervisor: Dr. Liu Guoxiang

- Built a comprehensive geological feature sample and knowledge base using multi-sensor, multi-temporal, and multi-regional datasets.
- Proposed a data- and knowledge-driven intelligent interpretation model for geological features in wide-area mountainous regions.
- Developed methods for the intelligent extraction of geomorphology, lithological groups, and key railway-related facilities using fused data sources and geological rules.

- Advanced the geometric distortion correction of SAR imagery, achieving high-precision image rectification.
- Contributed to the development of a proprietary software system for efficient and accurate extraction of geological features in remote, high-relief terrain.
- **Focus areas:**
 - Dynamic geomorphological evolution and key facility extraction along the Sichuan–Tibet Railway
 - Hyperspectral remote sensing for lithological structure recognition in alpine valleys
 - Knowledge base construction and fine-scale reconstruction of typical geological sections

Analysis of Influencing Factors of Thermal Melting Collapses in Different Basins of the Qinghai-Tibet Plateau Based on Multi-Source Remote Sensing and Geospatial Analysis

Research Assistant | Jul 2025 – Sept 2025

Supervisor: Dr. Liu Lin

Affiliation: Department of Earth and Environmental Sciences, The Chinese University of Hong Kong

- Focus on thermal melting collapse disasters in different basins of the Qinghai-Tibet Plateau, and conduct systematic analysis of influencing factors based on multi-source remote sensing data, with emphasis on solving the problems of difficult identification and low quantification accuracy of disaster-driven factors under high-altitude and complex underlying surface conditions.
- Integrate nearly 10 years of time-series optical images (Landsat-8/9, Sentinel-2), multi-band SAR data (Sentinel-1, ALOS-2 PALSAR), and auxiliary geospatial data (Digital Elevation Model (DEM), soil type maps, meteorological observation data) to construct a basic database of thermal melting collapses covering typical basins, and realize the correlation matching between the temporal and spatial distribution of disasters and potential influencing factors.
- Develop a quantitative model for influencing factors that integrates remote sensing features and geographical parameters, focusing on analyzing the driving effects of factors such as temperature rise, precipitation change, topographic slope, and permafrost type on the occurrence frequency and scale of thermal melting collapses, and clarify the contribution rate of each factor and the mechanism of interactive influence.
- Use spatial analysis and machine learning methods to establish an evaluation index system for the susceptibility of thermal melting collapses in different basins of the Qinghai-Tibet Plateau, generate refined susceptibility zoning maps, and provide a scientific basis for disaster risk early warning and prevention.

PROFESSIONAL EXPERIENCE

Shanxi Coking Coal Group Co., Ltd.

UAV Operator and Data Analyst Intern | Jan 2025 – Feb 2025

- Executed UAV-based inspections over mining slopes and tailings dams in complex terrains and adverse weather conditions.
- Conducted flight planning, monitoring, and emergency return operations for high-quality data capture.
- Processed aerial data for 3D modeling and spatial analysis to support mine safety monitoring.
- Participated in safety management discussions and proposed improvements such as increased inspection frequency and early warning systems.
- Strengthened UAV operation skills and deepened understanding of automated inspection workflows in industrial safety contexts.

Ministry of Natural Resources - Third Institute of Aerial Surveying and Remote Sensing

Intern – Aerial Emergency Monitoring Center | Jul 2024 – Aug 2024

- Processed UAV-acquired images for 3D modeling and texture mapping of roads and buildings in Jiaxing, Zhejiang Province.
- Applied photogrammetric and GIS techniques to generate detailed visualizations for landscape planning.
- Gained hands-on experience in UAV data correction, stitching, and model refinement using remote sensing software tools.

- Collaborated with engineers and analysts to ensure modeling accuracy and visual realism, emphasizing detail control and cross-functional communication.
- ❑ **Shanghai Jiaotong University John Hopcroft Center**
Image Generation Inter | Sept. 2025 – Present
- Led the improvement of core modules in image generation algorithms. Addressed the sampling efficiency issue of the Stable Diffusion model by optimizing scheduler logic and attention mechanisms.
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TECHNICAL SKILLS

Programming Languages: Python, C++, C, C#

Data Analysis: MATLAB

Remote Sensing & GIS:

- SAR Image Correction and Processing
- Optical/SAR Data Fusion
- Deep Learning for Feature Extraction
- UAV Image post-processing and 3D Reconstruction
- Visual Modeling and Texture Mapping
- Machine learning for Feature Extraction